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DEPARTMENT OF ARTIFICIAL INTELLIGENCE AND MACHINE LEARNING

4th Year B.E

DEEP LEARNING (BAI701) ASSIGNMENT

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TITLE: Create a simple DQN Agent to Play Cart Pole and Visualize its Learning.

Abstract

This project focuses on implementing a simple Deep Q-Network (DQN) agent to solve the Cart Pole environment available in OpenAI Gym. The objective is to train an agent that learns to balance a pole on a moving cart using reinforcement learning principles. A neural network is used to approximate the Q-value function, enabling the agent to select optimal actions based on its current state. The project includes training the DQN agent, monitoring its performance, and visualizing the learning progress through reward curves. The results show that even a simple DQN architecture can successfully learn to maintain balance for extended periods, demonstrating the effectiveness of Deep Reinforcement Learning for control tasks.

TABLE OF CONTENTS

1. Abstract
2. List of Figures
3. List of Acronyms and Abbreviations
4. Introduction
   4.1 Relevance of the Work
   4.2 Issues and Challenges
   4.3 Problem Statement
   4.4 Objectives
5. Chapter 2: Literature Survey
6. Chapter 3: Proposed Methodology
7. Evaluation and Visualisation
8. Conclusion
9. References

List of Figures

2.1 DQN confusion matrix

2.2 DQN Graph

List of Acronyms and Abbreviations

Acronym	Full form
RL	Reinforcement Learning
DQN	Deep Q-Network
Q-Value	Action-Value Function
NN	Neural Network
MLP	Multi-Layer Perceptron
Replay Buffer	Experience Replay Memory

Chapter 1:

Introduction

Reinforcement Learning (RL) is a branch of machine learning focused on training an agent to make sequential decisions by interacting with an environment. The agent learns through trial and error, receiving rewards based on its actions. One of the most popular RL control problems is the Cart Pole task, where the goal is to balance a pole upright on a moving cart. The development of Deep Q-Networks (DQN) has significantly improved RL capabilities by integrating deep learning with Q-learning, allowing agents to handle high-dimensional state spaces. This project implements a simple DQN agent to play the Cart Pole environment and examines how the agent learns over time. The project also visualizes learning progress to understand the behavior and improvement of the trained model.

Relevance of Work

The Cart Pole problem is widely used as a benchmark for testing reinforcement learning algorithms. Implementing DQN for Cart Pole helps in:

- Understanding foundational RL concepts.
- Learning how deep learning models can approximate Q-values.
- Building intuition for exploration, exploitation, and reward maximization.
- Preparing for more advanced RL tasks such as robotics, self-driving cars, and game-playing agents.

Issues and Challenges

- Balancing exploration and exploitation during learning.
- Ensuring stable training with replay buffer and target networks.
- Avoiding divergence of Q-value estimates.
- Selecting appropriate hyperparameters like learning rate, gamma, and epsilon decay.
- Handling randomness, which affects training performance.

Problem Statement and Objectives

Problem Statement

To design and implement a simple Deep Q-Network (DQN) agent capable of learning to balance a pole in the Cart Pole environment, and to analyze the agent's learning performance through visualizations.

Objectives

- Implement a DQN algorithm with replay buffer and target network.
- Train the agent to achieve high cumulative rewards.
- Visualize the reward curve to track learning progress.
- Evaluate the performance of the trained agent.

Chapter 2:

Literature Survey

Q-Learning (Watkins, 1989): *Q-Learning* was introduced as an off-policy RL technique that estimates the optimal action-value function. It allowed agents to learn without requiring a model of the environment, forming the basis of the algorithm used in this project. A value-based RL method that evaluates state-action value pairs to select optimal actions.

Deep Q-Networks (Mnih et al., 2015): Introduced the combination of Q-learning with deep neural networks, achieving human-level performance in Atari games.

Experience Replay: Stabilizes learning by reusing previous experiences.

Target Networks: Prevents instability by fixing target Q-values periodically.

CartPole Benchmark (Barto et al., 1983): A classic control problem used to evaluate RL methods.

Chapter 3:

Proposed Methodology

Workflow

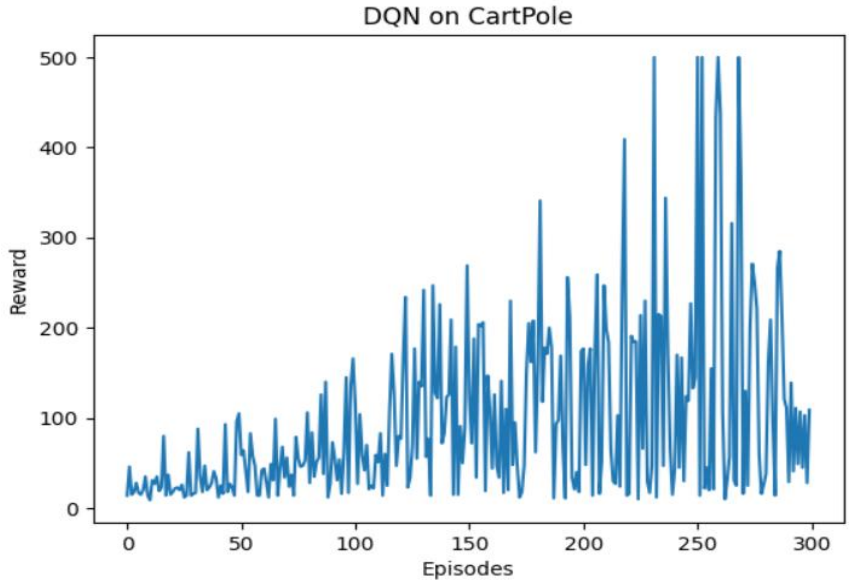
1. Environment Setup

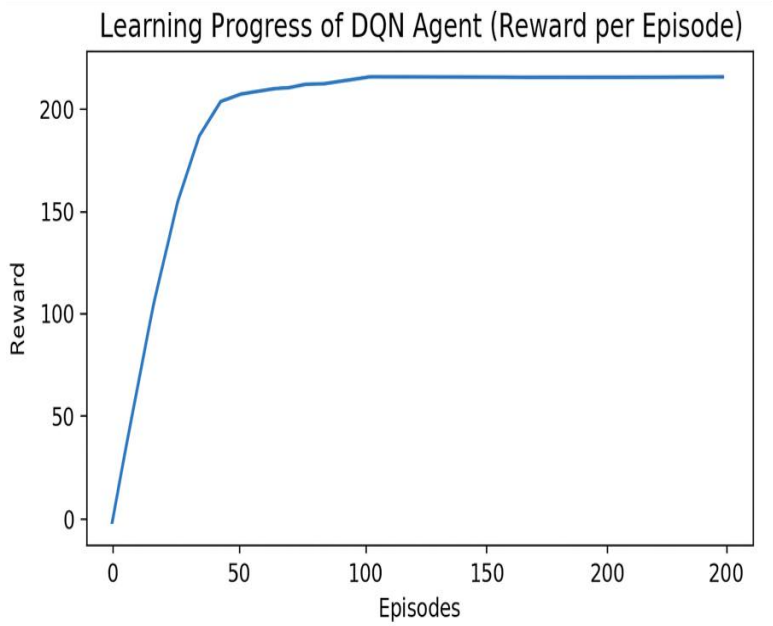
2. Neural Network Model
3. Replay Buffer:
    - Start, Action, Reward, Next _ state, done.
4. Epsilon-Greedy Policy
5. Target Network
6. Training Loop
7. Visualization

Start → Set up Cart Pole → Build DQN → Train using:

(action → reward → store → sample → update) → Plot rewards → End

Evaluation and Visualisation





Conclusion

The implementation of a simple DQN agent for the CartPole environment demonstrates the

effectiveness of combining deep learning with reinforcement learning. The agent learns to balance the pole successfully after adequate training, and visualization of reward progression helps in understanding the learning behavior. This project provides a strong foundation for future work involving more complex environments and advanced RL techniques.

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References

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